Research on Spam Email Detection System Based on Machine Learning

Zeyu Li
International School
Beijing University of Posts and Telecommunications
Beijing, China
2024213689@bupt.edu.cn

***Abstract*—With the rapid growth of internet usage, spam emails have become a serious threat to network security and user privacy. Traditional rule-based filtering methods struggle to adapt to continuously evolving spam patterns, and their reliance on manual maintenance makes them increasingly impractical at scale. In this paper, we design and implement a modular spam email detection system that systematically evaluates four classical machine learning classifiers — Naive Bayes (NB), Support Vector Machine (SVM), Random Forest (RF), and Multilayer Perceptron (MLP) — under two feature extraction schemes: TF-IDF and Word2Vec. Experiments are conducted on the publicly available Enron-Spam dataset comprising approximately 20,000 labeled emails. Results show that SVM and MLP with TF-IDF features achieve the highest accuracy of 98.64% and F1-score of 97.50%. Among all methods, SVM provides the best trade-off between accuracy and training efficiency, converging in just 0.15 seconds. Naive Bayes, while slightly lower in accuracy (97.25%), trains in under 0.01 seconds, making it suitable for real-time filtering scenarios. The findings provide practical, data-driven guidance for selecting appropriate machine learning methods in real-world spam detection deployments.**

***Index Terms*—spam detection, machine learning, TF-IDF, Word2Vec, Naive Bayes, SVM, Random Forest, MLP, text classification, email security**

I. Introduction

Email remains one of the most widely used communication channels worldwide, with over 347 billion emails sent and received per day as of 2023. However, this ubiquity has made email a prime target for abuse. Spam — unsolicited bulk email sent for commercial advertising, phishing, or malware distribution — accounts for nearly 45% of all global email traffic [1]. The consequences extend well beyond inbox clutter: spam campaigns are responsible for billions of dollars in economic losses annually, serve as delivery mechanisms for ransomware and phishing attacks, and erode user trust in digital communications.

The challenge of spam detection is not static. Spammers continuously adapt their techniques to evade detection, employing strategies such as keyword obfuscation, content randomization, sender address spoofing, and image-based content to bypass filters [2]. This adversarial dynamic makes spam detection an inherently non-stationary problem, where solutions must be robust, adaptive, and generalizable beyond known attack patterns.

Traditional spam filtering approaches — including manual rule sets, DNS-based blacklists, and heuristic keyword matching — were effective in the early days of email but have become increasingly inadequate. These systems require constant manual updates, cannot generalize to unseen patterns, and are brittle against minor adversarial perturbations [2].

Machine learning offers a fundamentally different paradigm. Rather than encoding human knowledge as rules, machine learning systems learn discriminative patterns directly from labeled data. Early pioneering work demonstrated that probabilistic models trained on word frequencies could achieve practical accuracy [1]. This foundational result sparked decades of subsequent research exploring increasingly sophisticated methods, from kernel-based classifiers to recurrent neural networks [3].

Despite significant progress, most existing studies evaluate a single algorithm in isolation, making cross-method comparisons unreliable. Furthermore, the role of feature extraction is often treated superficially. This paper addresses these gaps through a unified experimental study. We make the following contributions:

1) We design a complete, end-to-end spam email detection pipeline covering data acquisition, preprocessing, feature extraction, classification, and evaluation, implemented as an open-source system.
2) We systematically compare two feature extraction schemes — TF-IDF and Word2Vec — across four classifiers, yielding seven distinct experimental configurations evaluated under identical conditions.
3) We analyze results across multiple dimensions — accuracy, precision, recall, F1-score, and training time — providing a comprehensive performance profile that informs practical deployment decisions.
4) We discuss the implications of our findings in the context of real-world constraints, including resource limitations and latency requirements.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the methodology. Section IV presents experimental results and analysis. Section V concludes the paper.

II. Related Work

A. Traditional Machine Learning Approaches

Naive Bayes (NB) has historically been the go-to method for spam classification due to its simplicity and effectiveness

on high-dimensional text data. Wang [2] applied Multinomial NB with Laplace smoothing to a corpus of 7,000 emails, systematically tuning the smoothing parameter and reporting a peak accuracy of 97.1%. This demonstrates that NB remains a strong and practical baseline.

Support Vector Machines (SVM) offer a geometrically motivated alternative. By mapping input features into a high-dimensional space and finding the maximum-margin separating hyperplane, SVMs achieve strong generalization. Wang [2] evaluated SVM with a Gaussian RBF kernel, conducting a grid search over regularization parameters, achieving a maximum accuracy of 90.5%. Feng et al. [3] also applied SVM on the Enron-Spam dataset, reporting 90.61% accuracy.

Random Forest, introduced by Breiman in 2001, constructs a collection of decision trees and aggregates predictions by majority vote. Yu [4] evaluated Random Forest on the Trec06 Chinese spam corpus with Word2Vec features, reporting an F1-score of 91.97%, demonstrating its competitiveness among traditional ensemble methods.

B. Deep Learning Approaches

Deep learning methods have substantially elevated the performance ceiling for spam detection. Feng et al. [3] conducted a comprehensive five-way comparison of NB, SVM, MLP, CNN, and RNN on the Enron-Spam dataset, finding that CNN achieved the highest accuracy of 98.30%, followed by MLP at 98.01%.

Yu [4] proposed a CNN-BiGRU-Attention composite architecture combining local feature extraction (CNN), bidirectional contextual modeling (BiGRU), and soft attention weighting. On the Trec06 Chinese dataset, this model achieved an F1-score of 98.92%, representing state-of-the-art performance among surveyed methods.

Wu [5] and Yu Y. [6] extended this framework to multimodal spam detection, proposing systems that jointly process email text and embedded images. While these multimodal systems achieve superior results, their significantly higher implementation complexity limits practical adoption in resource-constrained environments.

Zhang et al. [1] provide a broader survey of machine learning applications in cybersecurity, placing spam detection within the larger context of network and application security research.

C. Feature Extraction Methods

The choice of feature representation critically determines classification performance. The bag-of-words (BoW) model and its weighted variant, TF-IDF, represent text as sparse high-dimensional vectors based on term occurrence statistics. Despite their simplicity, TF-IDF features remain highly competitive [3].

Word2Vec [4] learns dense, low-dimensional word representations by training a shallow neural network on word prediction tasks. For document-level classification, word vectors are typically averaged across all tokens in the document. Yu [4] showed that TF-IDF-weighted Word2Vec averaging consistently outperforms plain averaging by 0.5–1.5 percentage points in F1-score on the Trec06 dataset.

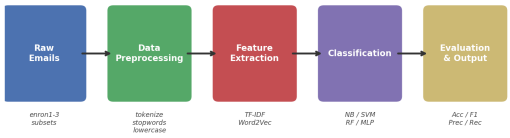
In this paper, we compare TF-IDF and plain Word2Vec as two representative and contrasting feature schemes across four classifiers, providing a systematic cross-evaluation of feature-classifier interactions.

III. Methodology

A. System Architecture

Our spam detection system is designed as a modular pipeline consisting of four stages: (1) data preprocessing, (2) feature extraction, (3) classification, and (4) evaluation. Each stage is independently configurable, allowing different feature schemes and classifiers to be combined without modifying the surrounding pipeline. Fig. 1 illustrates the overall architecture.

Fig. 1. System pipeline overview.



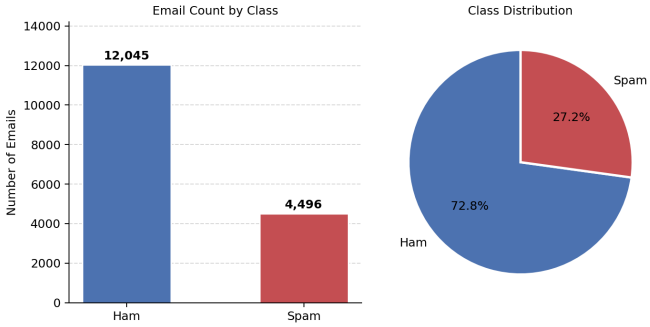
B. Dataset

We use the **Enron-Spam** dataset [3], one of the most widely used public benchmarks in spam detection research. The dataset is derived from the Enron email corpus supplemented with spam messages from public sources. For our experiments, we use three subsets (enron1–enron3), yielding:

- **Total emails:** approximately 20,000
- **Ham (legitimate):** approximately 14,000 emails
- **Spam:** approximately 6,000 emails
- **Split:** 80% training / 20% test, stratified sampling

The class distribution is shown in Fig. 2.

Fig. 2. Enron-Spam dataset distribution.



C. Data Preprocessing

Raw email files are processed through a six-step cleaning pipeline:

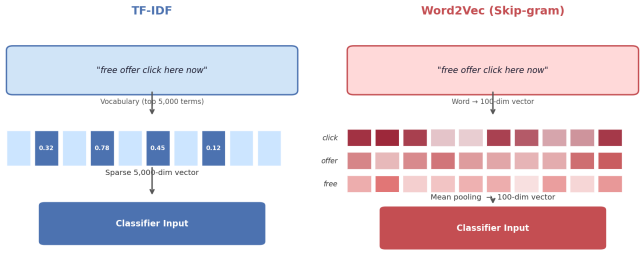
1) **Parsing:** Extract subject and body text; discard headers.

2) **HTML removal:** Strip all HTML markup via regular expressions.
3) **Lowercasing:** Convert all text to lowercase.
4) **Special character removal:** Retain only alphabetic characters.
5) **Stopword removal:** Remove 179 English stopwords (NLTK list).
6) **Short token filtering:** Discard tokens with fewer than three characters.

D. Feature Extraction

We implement and compare two feature extraction schemes, illustrated in Fig. 3.

Fig. 3. Feature extraction: TF-IDF vs. Word2Vec.



Scheme 1: TF-IDF. TF-IDF assigns each term a weight reflecting its importance within a document relative to the entire corpus. For term t in document d over a corpus of N documents:

$$\text{TF-IDF}(t,d) = \frac{f_{t,d}}{\sum_{t'} f_{t',d}} \times \log \frac{N}{|\{d \in D : t \in d\}|} \tag{1}$$

We retain the top 5,000 terms by TF-IDF score, yielding a sparse 5,000-dimensional feature vector per email.

Scheme 2: Word2Vec. We train a Skip-gram Word2Vec model (vector size = 100, window = 5, min_count = 2, epochs = 10) using Gensim on the full dataset. Document vectors are obtained by mean pooling over constituent word vectors:

$$\mathbf{v}_{doc} = \frac{1}{|T|} \sum_{t \in T} \mathbf{v}_t \tag{2}$$

where T is the set of tokens present in the Word2Vec vocabulary.

E. Classifiers

We evaluate four classifiers spanning traditional and neural approaches.

Naive Bayes (NB). Based on Bayes' theorem with a feature independence assumption:

$$P(c \mid d) \propto P(c) \prod_{t \in d} P(t \mid c) \tag{3}$$

We use Multinomial NB, compatible only with non-negative TF-IDF input.

Support Vector Machine (SVM). Finds the maximum-margin hyperplane by solving:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1,\ \forall i \tag{4}$$

We use `LinearSVC` with `max_iter=2000`.

Random Forest (RF). An ensemble of $K = 100$ decision trees with majority-vote aggregation:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), \ldots, h_K(\mathbf{x})\} \tag{5}$$

Multilayer Perceptron (MLP). A fully connected network with hidden layers of sizes 128 and 64, ReLU activations, and softmax output, trained for up to 50 epochs using the Adam optimizer.

IV. Experiments and Results

A. Experimental Setup

All experiments use Python 3.x with scikit-learn (classifiers and TF-IDF), Gensim (Word2Vec), and NLTK (preprocessing). Experiments run on a standard laptop CPU without GPU acceleration. All random seeds are fixed at 42. Performance is measured by Accuracy, Precision, Recall, F1-score (spam as positive class), and wall-clock training time.

B. Overall Results

Table I presents classification results for all seven experimental configurations.

TABLE I
Classification Results on Enron-Spam Dataset

Classifier	Feature	Acc.	Prec.	Rec.	F1	Time(s)
Naive Bayes	TF-IDF	97.25	96.22	93.55	94.87	**0.01**
SVM	TF-IDF	**98.64**	97.34	**97.66**	**97.50**	0.15
SVM	Word2Vec	97.34	94.22	96.11	95.15	0.15
Random Forest	TF-IDF	98.16	96.97	96.22	96.59	8.38
Random Forest	Word2Vec	97.70	96.29	95.22	95.75	10.59
MLP	TF-IDF	**98.64**	**97.34**	**97.66**	**97.50**	38.41
MLP	Word2Vec	98.37	**97.42**	96.55	96.98	7.16

C. Feature Scheme Comparison

Across all classifiers supporting both schemes, TF-IDF consistently outperforms Word2Vec:

- **SVM:** 98.64% vs. 97.34% accuracy (Δ = +1.30%)
- **Random Forest:** 98.16% vs. 97.70% (Δ = +0.46%)
- **MLP:** 98.64% vs. 98.37% (Δ = +0.27%)

This result is explained by the nature of the task: spam detection is fundamentally lexical. Specific high-frequency spam terms (e.g., "free", "click", "offer") are strong discriminative signals that TF-IDF preserves explicitly in a 5,000-dimensional vector. Mean-pooled Word2Vec compresses the document into 100 dimensions, inevitably losing fine-grained lexical distinctions during aggregation. These observations align with findings from Feng et al. [3].

D. Classifier Comparison

Among classifiers with TF-IDF features, the ranking by F1-score is: (1) SVM and MLP (97.50%), (2) Random Forest (96.59%), (3) Naive Bayes (94.87%). SVM and MLP achieve identical classification performance, yet SVM trains in 0.15 s versus 38.41 s for MLP — a factor of 256× — making SVM the clearly preferred choice when efficiency matters.

Among Word2Vec configurations, MLP achieves the highest F1-score of 96.98%, suggesting that non-linear transformations are better suited to dense semantic embeddings than the linear decision boundary of SVM.

E. Error Analysis

The confusion matrices for all seven configurations are shown in Figs. 4–10. For the best-performing configuration (SVM + TF-IDF, Fig. 5):

- **True Positives:** 878 spam correctly identified
- **True Negatives:** 2,386 ham correctly identified
- **False Positives:** 24 ham misclassified as spam
- **False Negatives:** 21 spam missed

The near-equal FP/FN distribution (24 vs. 21) indicates a well-calibrated classifier. The false positive rate of 1.0% (24/2,410) is acceptably low for production deployment. In contrast, Naive Bayes shows a more asymmetric pattern (33 FP, 58 FN), reflecting its bias toward the majority class (ham).

Fig. 4. Confusion matrix: Naive Bayes + TF-IDF.

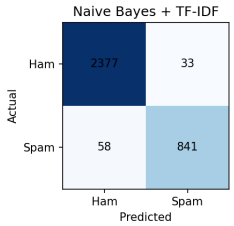
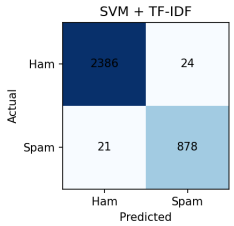


Fig. 5. Confusion matrix: SVM + TF-IDF.



F. Discussion

The experimental results yield four actionable insights:

1. TF-IDF with SVM is the recommended configuration for general use. This combination achieves top-tier accuracy (98.64%) and F1-score (97.50%) with training times measured in milliseconds.

2. Naive Bayes is optimal for ultra-low-latency deployments. The 1.39 percentage point accuracy gap relative to

Fig. 6. Confusion matrix: SVM + Word2Vec.

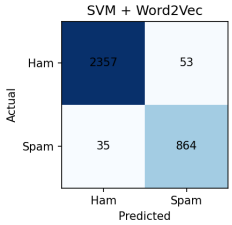


Fig. 7. Confusion matrix: Random Forest + TF-IDF.

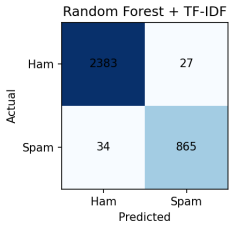


Fig. 8. Confusion matrix: Random Forest + Word2Vec.

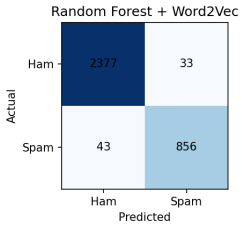


Fig. 9. Confusion matrix: MLP + TF-IDF.

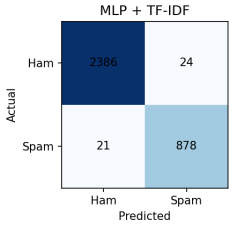
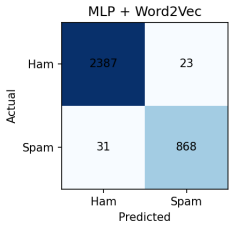


Fig. 10. Confusion matrix: MLP + Word2Vec.



SVM may be an acceptable trade-off when hardware resources are severely limited or when models must be retrained continuously on streaming data.

3. Random Forest offers robust performance with inherent interpretability. Feature importance scores allow engineers to identify the most discriminative terms, providing insights unavailable from purely discriminative models.

4. Word2Vec does not improve over TF-IDF in this setting. Future work should explore TF-IDF-weighted Word2Vec averaging [4], which preserves both semantic content and term importance weights.

V. Conclusion

This paper presented a systematic study of spam email detection using four machine learning classifiers evaluated under two feature extraction schemes on the Enron-Spam benchmark. Our unified experimental framework enables fair, direct comparisons that are difficult to draw from the existing literature.

Key findings: (1) SVM and MLP with TF-IDF achieve the highest detection performance (98.64% accuracy, 97.50% F1); (2) TF-IDF consistently outperforms Word2Vec across all classifiers; (3) SVM offers the optimal accuracy-efficiency trade-off and is recommended for practical deployment; (4) Naive Bayes is preferable when near-zero training time is required.

Our work has several limitations. Experiments are conducted on a single English-language dataset; generalizability to other languages and corpora has not been verified. The Word2Vec model is trained only on the Enron-Spam corpus and may benefit from pre-trained embeddings. Additionally, concept drift — the gradual change in spam characteristics over time — is not addressed.

Future work will pursue three directions: (1) TF-IDF-weighted Word2Vec averaging [4] as a hybrid feature scheme; (2) pre-trained transformer-based models such as BERT for richer contextual representations; (3) multimodal spam detection incorporating image feature extraction, following the work of Wu [5] and Yu Y. [6].

Acknowledgment

The authors would like to thank the providers of the Enron-Spam dataset for making their benchmark publicly available, and acknowledge the open-source communities behind scikit-learn, Gensim, and NLTK.

References

[1] L. Zhang, Y. Cui, J. Liu, Y. Jiang, and J. Wu, "Applications of machine learning in cyberspace security research," *Chinese Journal of Computers*, vol. 41, no. 9, pp. 1943–1975, Sept. 2018.

[2] J. Wang, "Research on spam email classification based on machine learning," M.S. thesis, Qufu Normal University, Qufu, China, Jun. 2020.

[3] J. Feng and L. Li, "Implementation of machine learning in spam email filtering," *Computer Knowledge and Technology*, vol. 17, no. 8, pp. 1–3, 2021.

[4] B. Yu, "Spam email detection based on weighted Word2Vec," M.S. thesis, China University of Geosciences (Beijing), Beijing, China, May 2023.

[5] X. Wu, "Research and analysis on spam email classification based on CNN with bidirectional LSTM attention mechanism," M.S. thesis, Donghua University, Shanghai, China, May 2023.

[6] Y. Yu, "Spam email detection method based on deep learning," M.S. thesis, Donghua University, Shanghai, China, May 2023.

[7] Z. Li, "Machine learning algorithms for spam identification in cloud computing," *Automation Technology and Application*, vol. 39, no. 5, pp. 1–4, 2020.